

REVIEW

A systematic literature review on the application of artificial intelligence techniques for rock strength estimation

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Abstract

This paper presents a systematic literature review on the prediction of unconfined compressive strength (UCS) and elastic modulus (E) with artificial intelligence (AI) models. The study categorises three essential parts: (1) a combination of physical and mechanical properties, (2) mechanical properties, and (3) physical properties as input parameters for AI models in estimating UCS and E. The review selection was based on search keywords using title-abstract, full-text, and keywords from Scopus and Web of Science online database libraries. A total of 131 peer-reviewed research articles published from 2014 to 2024 were critically reviewed to provide answers to research-related questions related to current advancements in the prediction of UCS and E with AI models. Among the AI technologies analysed, artificial neural networks (ANN) and ANN-based models stand out as the most used AI algorithms; other algorithms, including ANFIS, RF, SVM, and XGBoost model, have been used at significant levels in predicting UCS and E with high prediction accuracy of R^2 greater 0.90 with minimum mean error margins. The ANN (24.7%), ANFIS (11.7%), and RF (7.6%) have been essentially employed in many research studies to predict rock strength. The study combined mechanical and physical properties with AI models at approximately 59%, and after that, mechanical properties at 23.6%. The efficiency of AI algorithms and their application is associated with the usage of data and input parameters. This review recommends future study gaps and places emphasis on integrating rock mechanics, physical laws (Mohr–Coulomb and Hoek–Brown failure criteria) and adaptive AI techniques to advance the adaptability and reliability in predicting rock strength and deformation characteristics.

Keywords Compressive strength · Artificial intelligence · Rocks · Elastic modulus · Physical–mechanical properties · Prediction

Abbreviations

AI	Artificial intelligence
ANFIS	Adaptive neural fuzzy inference system
ANN	Artificial neural network
ML	Machine learning
DL	Deep learning
UCS	Unconfined compressive strength
UCT	Unconfined compressive test
RF	Random forest
SVM	Support vector machine



ICA	Imperialist competitive algorithm
XGBoost	Extreme gradient boost
FIS	Fuzzy inference system
ELM	Extreme learning machine
CNN	Convolutional neural network
DNN	Deep neural network
RNN	Recurrent neural network
MLP	Multi-layer perceptron
GEP	Gene expression programming
LSSVM	Least square support vector machine
CART	Classification and regression tree
KNN	K-nearest neighbours
MLR	Multiple linear regression
COA	Cuckoo optimisation algorithm
GP	Gaussian process
MPMR	Minimax probability machine regression
BPNN	Back propagation neural network
SANN	Subtree-based attention neural network
GA	Genetic algorithm
RMSE	Root mean square error
MSE	Mean square error
CFNN	Cascaded forward neural network
AAPE	Average absolute percentage error
LMA	Levenberg–Marquardt algorithm
BPNN	Back propagation neural network
LOG	Laplacian of Gaussian
MELM	Multi-layer extreme learning machine
COA	Cuckoo optimisation algorithm
TS	Tensile strength
BTI	Brazilian tensile index
BPI	Block punch index
PSO	Particle swarm optimisation
SLR	Systematic literature review
E	Elastic modulus
GFFN	Generalised feedforward network
MAPE	Mean average percentage error
GPR	Gaussian process regression
GBDT	Gradient boost decision tree
OB	Bayesian optimisation
SCM	Subtractive clustering method
MAPE	Mean absolute percentage error
M5T	Tree-based machine learning
GWO	Grey Wolf optimiser
KELM	Kernal extreme learning machine
SSA	Singular spectrum analysis
RVM	Relevance vector machine
MVRA	Multivariable regression analysis
VAF	Variance account for

MAE Mean average error

List of symbols

V_p	P-wave velocity
R_n	Schmidt hammer rebound number
$I_{s(50)}$	Point load index
BPI	Block punch index
ρ_d	Dry density
n	Porosity
Qtz	Quartz
Plg	Plagioclase
ρ	Density
e	Poisson ratio
m_c	Moisture content
A_b	Water absorption
v_d	Dynamic poisson ratio
V_s	S-wave velocity
ρ_s	Saturates density
β	Serpentinisation percentage
γ_d	Dry unit weight
γ_s	Saturated unit weight
E_d	Dynamic Young's modulus
n_e	Effective porosity
μ_d	Dynamic Poisson ratio
Fr	Feldspar
Fg	Fragments
n_t	Total porosity
C_t	Compression time
S_t	Shear time
Kpr	Alkali feldspar
Chl	Chlorite
γ	Dry unit weight

1 Introduction

Rock strength prediction is vital in many industries for optimising operations, ensuring stability and safety, and reducing project costs. The application of rock strength ranges from civil and mining engineering to energy production and environmental management [1, 2]. The accurate unconfined compressive strength (UCS) and elastic modulus (E) of rock data offer information for decision-making in designing efficient and safe projects to minimise risk to human life and the environment. By integrating more advanced prediction techniques, such as artificial intelligence (AI) models, into rock strength assessment, it is possible to enhance the prediction accuracy and reliability of rock strength properties, including UCS and E, profiting from a vast range of real-world applications.

The rock strength parameter, such as UCS, is a significant parameter for design and stability analysis, especially when rocks are under compressive stresses [3], and it can be obtained through a series of laboratory tests [4, 5]. The most common laboratory test is the unconfined compression test, UCT, which provides rock strength characteristics like uniaxial compressive strength, UCS, and elastic modulus, E. The UCT is a straightforward

method used to evaluate the UCS of rocks. Nonetheless, some challenges exist in determining UCS and E with UCT. For instance, acquiring a high-quality rock core is a challenge, notably for highly weathered and fragmented rock masses [6]. Additionally, UCT has several disadvantages, such as being expensive and time-consuming [7]. Therefore, several empirical models have been developed based on simple and multiple regression methods with different rock strength properties from direct and indirect test results to estimate UCS and E.

An indirect and direct measurement approach including P-wave velocity (V_p), density, Schmidt hammer rebound number (R_n), and point load index ($I_{s(50)}$) has been used due to the high cost of UCT equipment for predicting rock strength is more favourable and appropriate as these tests are economical and easy to carry out [8]. $I_{s(50)}$ [9, 10], R_n [11, 12], and V_p [13, 14] are widely utilised to predict UCS and E using simple and multiple regressions. The indirect and direct estimation techniques of UCS and E have been considered in the literature and developed several empirical models using linear simple regression analysis [7, 15], multiple regression methods [16, 17], and nonlinear regression analysis [18, 19] with a meaningful relationship between UCS and E. In the other studies by [20, 21], the correlation of indirect test parameters did not provide reliable values for UCS and E. For example, Mohamad et al. [22] studied the correlation between $I_{s(50)}$ and UCS, classifying the Penang Island granitic rock from weak (24.76 MPa) to very strong (156.82 MPa). Through simple and multiple regression methods, Minaeian and Ahangari [23] investigated UCS with indirect methods, including V_p and R_n to estimate the relationship of UCS. Using RMSE and VAF, they noticed 94.34 and 1.56 regarding V_p and UCS and 94.39 and 1.6 between UCS and R_n through a simple regression model. Meanwhile, through multiple regression analysis for UCS, 97.24 and 1.34 were achieved. However, different empirical models have been reported on estimating UCS and E; there is no conclusive generalised empirical equation for rocks [24]. The empirical models from most of the studies are site-specific and applicable to a particular area [25], with low accuracy [24], and empirical rock techniques involve experience, observation, and personal judgement [26]. In order to deal with these shortcomings of empirical techniques, AI methods are introduced for predicting UCS and E using direct and indirect test results.

In the past decade, many scholars have proposed AI algorithms capable of predicting UCS and E of rocks [7, 27, 28]. The AI algorithms commonly used in predicting UCS include ANN [29–31], ANFIS [32–36], RF [35, 36], XGBoost [37], and SVM [38]. Among the AI algorithms, ANN is the most widely used technique to predict rock UCS and E. For example, some scholars have used hybrid optimised algorithms, including ICA-ANN and PSO-ANN, to estimate UCS and E [15, 39, 40]. In addition, ANFIS and ANFIS-based subtractive clustering methods (ANFIS-SCM) have also been utilised to predict the UCS and E [32, 41, 42]. The above AI models were reported to demonstrate accurate predictions of UCS and E. Also, the AI models were noticed to perform better than the empirical models. The call for the use of AI models in predicting UCS and E of rock relies on the AI model's ability to deal with nonlinear, complex, and multidimensional data. AI models provide rapid and more accurate predictions, are cost-effective and adapt to new data. Additionally, AI models provide significant advantages in terms of scalability, flexibility, and efficiency, making AI techniques gain currency in geotechnical engineering and rock mechanics compared to traditional empirical methods.

Besides the AI algorithms mentioned above, numerous effective AI models have been adopted to forecast UCS and E, including XGBoost, SVM, and CNN. Liu et al. [43] estimated the UCS of sandstone, exploring the framework of boost trees, XGBoost, AdaBoostM, and CGBoost. Also, Sun et al. [44] innovatively used CNN and X-ray computer tomography methods to estimate rock UCS. Cao et al. [37] developed an algorithm based on XGBoost with firefly algorithm (XGBoost-FA) to sumise UCS and E, comparing the XGBoost-FA with RBFN XGBoost and SVM the proposed XGBoost-FA algorithm achieving R^2 of 0.99. To investigate the UCS, non-destructive tests like V_p , n_e , and R_n were used by Skentou et al. [28] with ANN-PSO, ANN-LM, and ANN-ICA algorithms and noticed R^2 of 0.96 for ANN-LM compared to the other algorithms. Jahed et al. [32] used the ANFIS algorithm to estimate UCS and E of Pahang-Selangor granitic rock from Malaysia adopting V_p , density, plagioclase, and quartz. They used V_p , density, plagioclase, and quartz as the input parameters for the ANFIS algorithm, reporting R^2 of about 0.99 and 1.0 for UCS and E, respectively.

There are a sizable number of the literature on estimations of rocks UCS and E applying AI techniques, both traditional and hybrid algorithms, including ANN, FIS, ELM, RF, BO, ANN-TLBO, PSO-LSSVM, and ANN-PSO [45–49]. Although AI methods have considerable advantages and promising features in rock mechanics and engineering, there are still limitations. A significant research gap exists in applying advanced and adaptive AI models in rock mechanics. Recent studies predominantly depend on long-established AI methods, with limited studies on promising deep and reinforcement AI algorithms, potentially improving modelling reliability and predictability. Additionally, there is a significant need to integrate AI algorithms with physical theories to enhance consistency and pragmatical understanding. Enhancing the robustness of AI-driven rock mechanics and engineering solutions can be achieved by addressing the gaps.

This paper conducts a systematic literature review (SLR) to thoroughly explore the usage of AI techniques in rock engineering solutions. This review covers relevant peer-reviewed research articles published from 2014 to 2024. The objective of this review is to provide answers to the questions presented in Table 1. The SLR will provide further research based on the usage of AI methods in calculating UCS and E. AI applications in estimating UCS and E will highlight the potentials, available methods, and challenges. The SLR will also fill an essential gap in the literature. Additionally, the SLR will identify key areas for future research prospects.

2 Material and methods

2.1 Synopsis of SLR procedures

The protocol of this SLR adhered to three categories: (1) planning stage, (2) execution, and (3) discussion stages, as illustrated in Fig. 1. The SLR approach offers a synopsis of a primary study on a specific research subject that scientifically identifies, to choose from, assesses, and synthesises all quality evidence pertinent to that research theme or topic [50]. The PRISMA standard protocols [51, 52] were followed to develop this SLR. PRISMA guidelines are a universally accepted method that provides a checklist that is strictly applied in this work. This technique enhanced the quality assurance of the revision process and ensured its replication.

3 Research article search

The systematic search of articles was conducted across two databases, specifically the most widely known libraries, Scopus and Web of Science (WoS) online library databases, as in Table 2. Table 2 summarises the keywords employed to search the articles from the two databases relevant to the research topic by applying the “Boolean operators.” The keyword combination in the table was applied using the two databases through Griffith University’s online library system. The keywords used helped to facilitate the identification of high-quality literature from WoS and Scopus, which was screened using the PRISMA protocol. In addition, the PRISMA

Table 1 Quality assessment criteria for initial articles search

Research objective	Number	Questions
Research problems for the planning review	Main	Which AI algorithms have been used in the literature to predict UCS and E?
Sub-questions	1	What techniques are included in AI?
	2	What are the features of available AI techniques?
	3	How can AI techniques be used to predict UCS and E?
	4	What are the projections and future research trends?

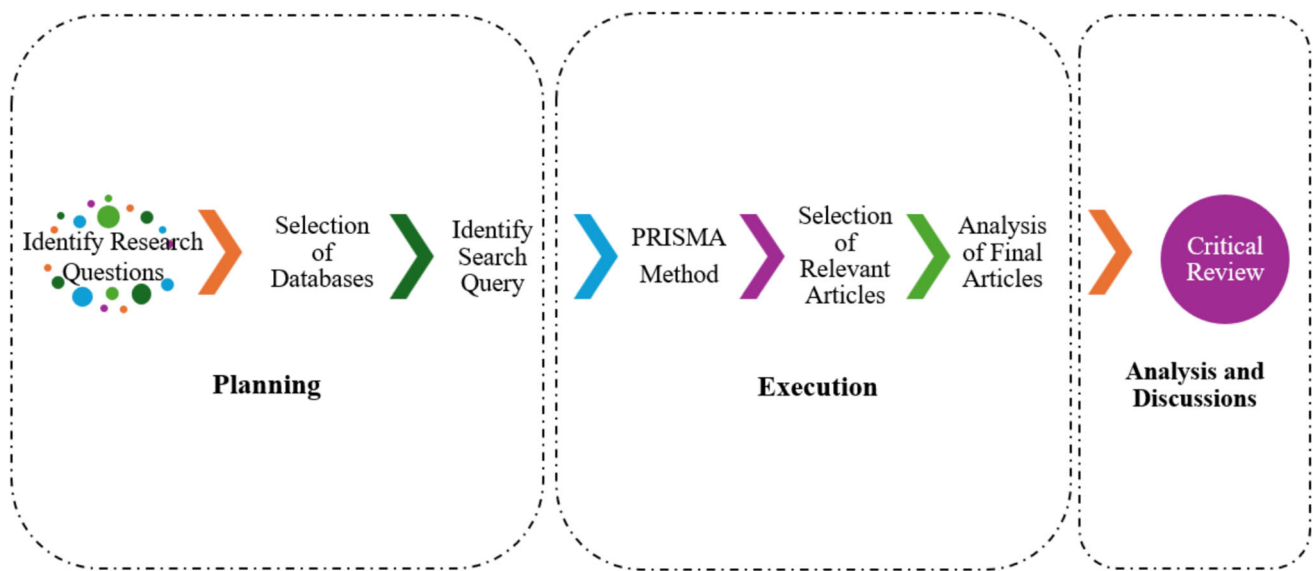


Fig. 1 The research process of this SLR

Table 2 Keywords, controllers, and early search outcomes from Scopus and WoS databases online library

Database	Search keys and strainers utilised	Search outcomes
Scopus	TITLE-ABS-KEY ((rock OR UCS OR AI OR ML OR DL) AND (SVM OR elastic modulus OR UCS OR ANN) AND (rock OR CNN OR UCS)) Refined By: Area of Study: Engineering; Publication Year: All Language: English	1763
WoS	AB = ((rock OR UCS OR AI OR ML OR DL) AND (SVM OR elastic modulus OR UCS OR ANN) AND (rock OR CNN OR UCS)) Refined By: Area of Study: Engineering; Publication Year: All Language: English	1503
WoS web of science		

technique provides an avenue to describe article collections and sorting, as in Table 2. It should be noted that only peer-reviewed journal articles published from 2014 to 2024 were selected for this paper.

3.1 Screening, exclusion, and inclusion criteria

A total of 3263 records were obtained from WoS and Scopus online library databases. In the initial screening processes, 750 duplicates were removed utilising the EndNote-21 reference manager tool. Next, 2513 documents were assessed in the EndNote-21 considering the title and abstract. Two hundred three (200) articles were recognised as hypothetically qualified after excluding 3063, as they fall outside the search criteria. The 200 articles (full text) were downloaded using the EndNote reference manager for further screening in Covidence software version 2.0.

Two reviewers independently screened each research article using Covidence software version 2.0, and the outcomes were assessed to confirm the impartiality of the review procedure using all criteria in Table 3. It was then followed by a thorough elimination of articles that did not have academic relevance and were unrelated to AI models in predicting UCS and E. After the covidence full-text screening, 97 full-text articles were excluded, as they fell out of the exclusion and inclusion criteria. Finally, 103 full-text articles were considered eligible for content analysis. The study excluded conference proceedings, book chapters, and grey literature, including (nonofficially published articles by a journal) as described in Sect. 2.5. The literature search included research

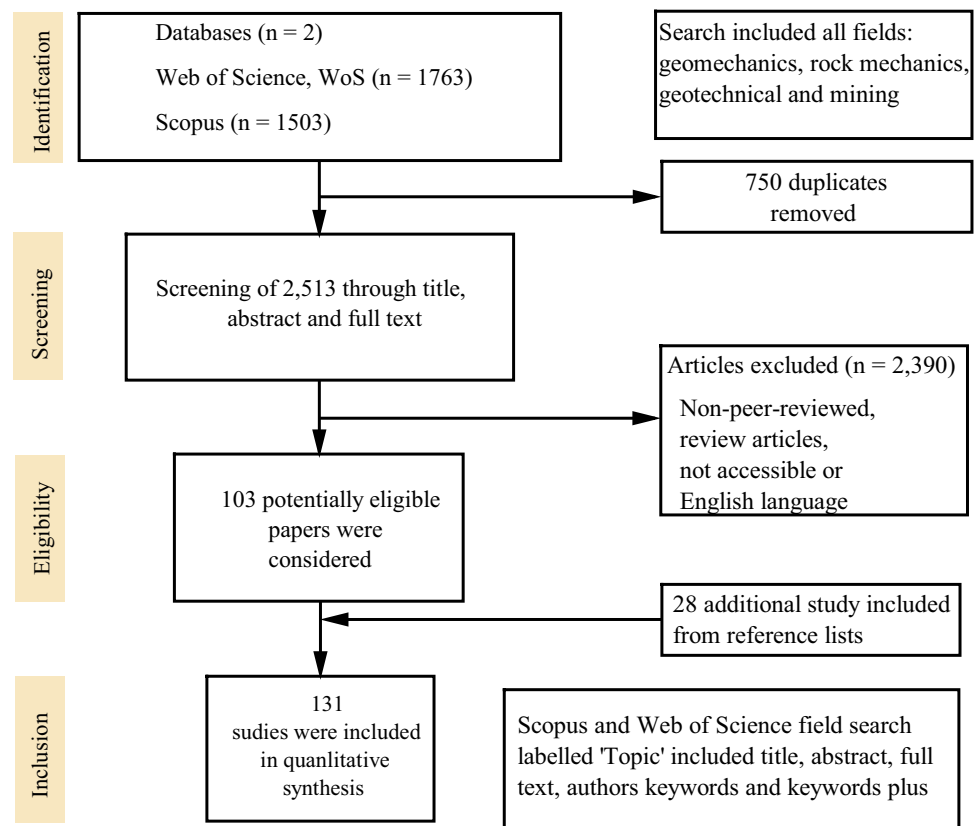
Table 3 Quality evaluation criteria

Number	Criteria	Inclusion/eligibility criteria	Exclusion criteria
1	Timeline	2014–2024	Before 2014
2	Literature type	Full research articles retrieved	The research articles that are not fully accessed
3	Topic	The topic relevant to the UCS, E and AI models	The topic that is not related to AI models, UCS and E
4	Peer-reviewed	Peer-reviewed scientific articles	Not peer-reviewed
5	Clear research content	Articles with explicit research titles, abstracts and keywords	Articles with no explicit research titles, abstracts and keywords
6	Rock type	Studies based on natural rocks only	Studies conducted on concrete blocks and rock-like material
7	Language	Articles in English language	Articles not in English language

articles published only in English to ease translation and avoid confusion. In addition, the English language is widely used in academic research, with a considerable number of published scholarly research articles and other relevant documents.

Further, 28 more records were included, which were obtained from reference lists in some of the articles included in this study. Ultimately, the full texts of 131 articles were studied. The assessment criteria and the SLR approaches were adopted to ensure neutrality and fairness during the literature selection. Figure 2 summarises the study procedure in the PRISMA flow chart. A SLR protocol was developed to outline the criteria for research article selection, record search plan, data retrieving, and analysis procedures, which are shown in Fig. 2. The search only included peer-reviewed literature to ensure scientific rigour. Grey literature, reviews, conference papers, and book chapters were intentionally excluded with the assumption that they were indirectly captured and expanded upon in scientific texts.

Fig. 2 Summary of SLR process: visualisation of adopted process from Prisma [53]



3.2 Data extraction

The 131 research articles after full-text screening in the Covidence software version 2.0 were extracted into Microsoft Excel 365, which includes the author(s) name, year of publication, keywords, and other relevant information on prediction UCS and E with AI algorithms, physical, and mechanical properties of natural rocks from direct and indirect tests results and the relevant model accuracy metrics in these studies. A meta-analysis was conducted based on the information acquired from the publications retrieved in this study. Each paper comprehensively reviewed the rock types and important AI model input parameters.

3.3 Studies limitations

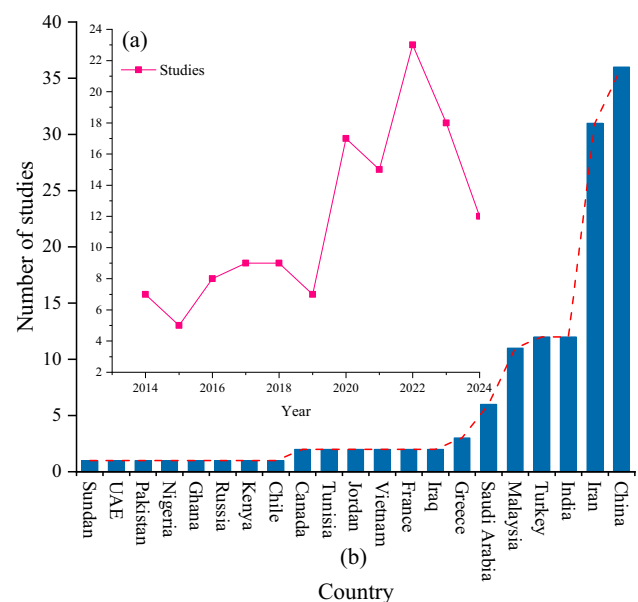
The primary limits of this study are (1) articles published in the English language that vetoed the inclusion of articles published in another language on the topic, the selection of English language only was to avoid translation, misinterpretation, misrepresentation of articles not in the English language, (2) the choice of a specific database skewed the search outcomes, and (3) the filtered keywords search approach may disregard some vital keywords. The restrictions may limit the realm of this review. However, efforts were put in place to alleviate possible preferences and guarantee the vigour of the outcomes via the adopted methodology. Furthermore, the SLR technique facilitates an utter check of the existing research articles, giving beneficial comprehension into AI, UCS, and E.

4 Results and discussion

4.1 Publication information and geographical context

Between the years of 2014 and 2024, all studies ($n = 131$) were peer-reviewed and published. Approximately 34% of research was produced from 2014 to 2019, after which most research ($n = 85$) was published in the last 5 years, as illustrated in Fig. 3(a). The most research articles ($n = 22$) were published in the 2022 fiscal year. As exhibited in Fig. 3(b), the majority of the studies were conducted in China ($n = 36$), Iran ($n = 31$), India and Turkey ($n = 12$ each), Malaysia ($n = 11$), Saudi Arabia and Greece ($n = 6$ and 3) each, 14 countries ($n = 1$ and

Fig. 3 **a** Growth of scientific production of research articles per year on rock UCS and AI; **b** geographical distribution of research articles' primary authors



2) articles each. Several studies conducted in some parts of the world are constrained. The column graph indicates that substantial efforts must be considered in diverse parts across the globe to institute the utilisation of AI algorithms in rock mechanics and engineering. It should be noted that many research articles were retrieved from 2023 to 2024 on studies involving the prediction of rock strength properties with AI models. However, per the research objectives and focus, only a few articles meet the criteria set for this study, and this is the reason there is a decrease in articles produced after 2022.

5 Keywords co-occurrence analysis

Figure 4 illustrates the co-occurrence of research keywords comprising computational methods with an AI algorithms skeleton for the last decade (2014–2024), including a total of 233 individual keywords, using VOSviewer software version 1.6.20. VOSviewer constructs a network and visualises bibliometrics based on the co-occurrence of keywords in the form of linked clusters and generates a cluster map. Each node in Fig. 4 corresponds to a keyword, with the node size implying occurrence frequency. The keyword size demonstrates the centrality and establishes the track of recognition in a research subject. Conspicuously, the crucial keywords in Fig. 4 are the adaptive neuro-fuzzy inference system, artificial neural networks, uniaxial compressive strength, rocks, rock properties, and elastic modulus. The linking lines joining the keywords imply the correlations, with the thick lines demonstrating the rate of occurrence.

By analysing the rate of keywords within the research domain, VOSviewer performed some statistical analysis to examine all the keywords listed in Table 4. The 10 keywords comprise the core ideas and associations in the domain of this review. Logically, the prevailing term is “artificial neural network.” Additionally, keywords like “uniaxial compressive strength,” “granite,” and “elastic modulus” indicate the deep focus on advances in AI algorithms and prediction models’ performance in recent studies on rocks. The existence of the “machine learning” keyword highlights the already extensive exploitation of machine learning technologies. Simultaneously, the presence of “nondestructive testing” and particle swarm optimisation keywords also highlights the use of indirect physical and mechanical properties in predicting UCS and E. The particle swarm optimisation demonstrates the exploration of hybrid AI algorithms to determine rock strength.

Fig. 4 Keywords and co-occurrence diagram

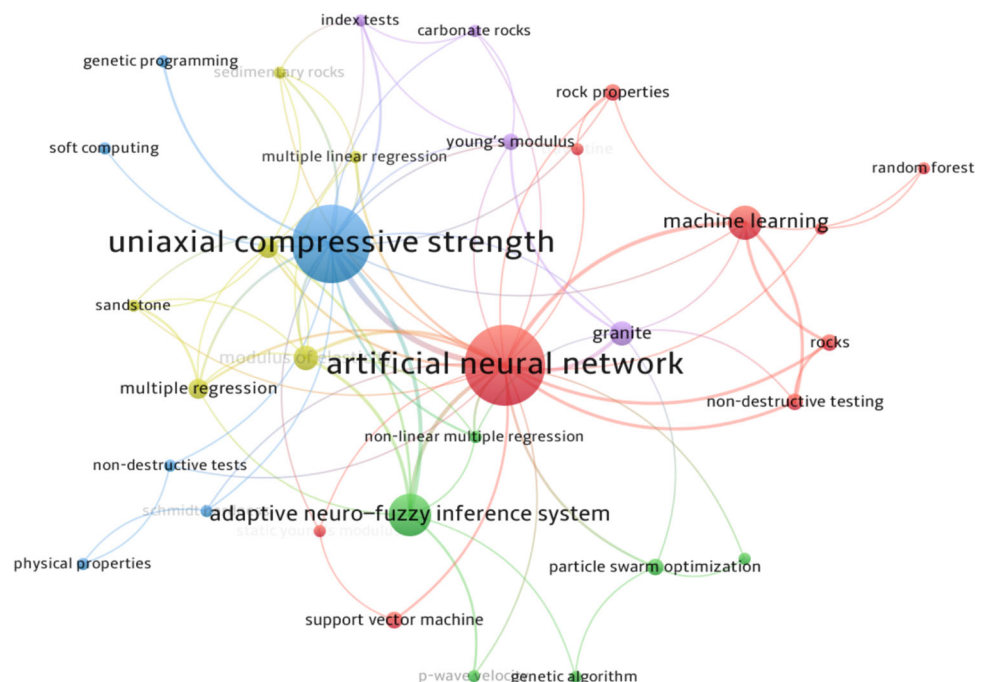


Table 4 Catalogue of occurrences and keywords obtained from 233 distinct themes

Keyword	Occurrence
Artificial neural network	26
Uniaxial compressive strength	25
Adaptive neuro-fuzzy inference system	11
Machine learning	8
Granite	5
Modulus of elasticity	5
Fuzzy inference system	4
Multiple regression	4
Nondestructive testing	3
Particle swarm optimisation	3

The illustrated cluster net map of computational mechanisms in AI frameworks is profound and satisfying, with discerning coherent connections of linking themes. The incorporation of AI models, like adaptive neuro-fuzzy inference systems, fuzzy inference systems, and hybrid particle swarm optimisation into the orthodox scope of rock mechanics, has advanced the prediction and evaluation of rock strength and deformation characteristics.

6 Synopsis of used AI methods

AI is a general canopy beneath which machine (ML) and deep (DL) learning fall. The overarching concept of AI encompasses creating machines capable of mimicking tasks typical of human intelligence. ML and DL provide advanced tools, including neural networks and their architecture, to enhance and execute complicated tasks. Numerous traditional learning models are available in the literature, for example, ANN, ANFIS, CNN, DNN, RF, SVR, SVM, BP, FIS, MLP, RNN, and LSTM. Among the AI models mentioned in the previous sections, ANN and ANN-based models are the most commonly used by researchers in the fields of rock mechanics and engineering and geotechnical engineering. An artificial neural network (ANN) effectively links input geotechnical and geometrical parameters, such as uniaxial compressive strength (UCS), density, porosity, Schmidt rebound hardness, Brazilian tensile strength, point load index, and P- and S-wave velocities to the desired output by learning from the training data.

The AI algorithms present unique features that make them appropriate for various applications in science and engineering. Table 5 summarises the advantages and limitations of some commonly used AI algorithms in the literature. Understanding each algorithm's advantages and limitations is essential for selecting an appropriate learning algorithm when facing a particular research task. Additionally, hybrid algorithms and ensemble learning approaches can be utilised effectively, integrating diverse learning algorithms for complex tasks to achieve robust accuracy. Furthermore, an integrated approach is especially beneficial for tackling complex problems that need multi-step solutions and involve diverse data types. However, using a combination of AI algorithms can achieve more effective results.

The employment of selecting suitable AI algorithms requires performance metrics and validation methods, which enable rigour accuracy assessment of efficient performance. Efficient and resource-efficient integration of AI algorithms in predicting UCS and E requires careful optimisation and adjustments due to limited resources and computational power. AI algorithm's effectiveness is typically evaluated using RMSE, accuracy, F1-score, precision, and recall metrics, as presented in the equations below [58]. The methods validate AI model reliability by testing them on diverse datasets, ensuring smooth integration into geotechnical engineering challenges through optimisation and adjustments.

Table 5 Advantages and limitations of some developed AI models

AI algorithm	Advantages	Limitations	References
ANN	Capable of predicting UCS with microfabric images, mainly from weak and highly stratified fractured rock	The problem is the opaqueness of the underlying relationship between inputs and outputs, often called the “black box.”	Ali et al. [54]
ANFIS	Best prediction with random sets of data	It requires broad ideas to establish relationships between different parameters and depends on membership functions	Umrao et al. [55]
RF	Minimal influence by hyperparameters, fast convergence speed, resistance to overfitting and noise insensitive to data, as well as best predictability	The predictive model is complex to explain physically and cannot reveal the mechanisms of the influence of input parameters on UCS	Wang et al. [36]
SVM	Provide sufficient accuracy for predicting intact rock UCS	Unsuitable for extensive data when faced with data noise and not capable of using a probabilistic explanation	Cemiloglu et al. [38]
DNN	Predict UCS and E with high accuracy	The algorithm is prone to overfitting due to the different number of parameters learned during the model building	Azarafza et al. [56]
FIS	It emerges as a competent model because of its efficiency in dealing with impreciseness and handling uncertainties in test data with high accuracy	It depends on membership function variables and fuzzy “if-then” rules involving user interference	Heidari et al. [57]
CNN	Shows superior performance for predicting UCS of rock with CT-scanned image datasets	It requires data from high-resolution X-ray scanning to calculate UCS	Sun et al. [44]

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - score = 2 \times \left(\frac{P \times R}{P + R} \right) \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_n - y_p)^2} \quad (5)$$

where TP: true positive, TN: true negative, FP: force positive, FN: force negative, P : precision, R : recall, y_n : is the measured value, y_p : is the mean of the measured value, and n is the absolute number of data.

7 Application of AI models in predicting UCS and E of rock

This section explores AI algorithms’ application in predicting UCS and E using physical and mechanical properties as documented in the literature. The first section presents the breakdown of subdomains typically used as input parameters and a brief statistical analysis of AI algorithms available in the literature on estimating UCS

and E. The three primary domains identified are (a) physical–mechanical properties, (b) mechanical properties, and (c) physical properties.

7.1 Breakdown of input parameters typically used for predicting UCS and E

Figure 5(a) provides the breakdown of the subdomain identified. Geotechnical materials like rocks have mechanical properties, including BTS, $I_{s(50)}$, V_p , BPI, modulus, and R_n , and physical properties such as density, porosity, quartz content, plagioclase, and particle size. It is clear from the figure that the majority of studies involve a combination of physical–mechanical properties as model input parameters ($n = 75$), followed by mechanical properties ($n = 33$) and then physical properties ($n = 23$) for predicting UCS and E. Figure 5(b) presents the percentage distribution of different physical and mechanical properties and the combination of physical and mechanical properties used as input parameters with AI algorithms in estimating UCS and E. As observed in Fig. 5(b), most studies utilised the combination of physical–mechanical parameters in predicting UCS and E, representing 57.3%. The reason could be that the incorporation of the combination of physical and mechanical properties of rock with AI models allows for better accuracy, holistic, and generalisation in the prediction of UCS and E. The combination of the two properties can help to capture the complex nature of the multi-facet of rock, which leads to enhanced performance in estimating the UCS and E of different rocks under various geological and environmental conditions and in diverse rock mechanics and geotechnical engineering applications. Additionally, the combination of physical and mechanical properties helps reduce the risk of AI models overfitting.

The combination of physical and mechanical properties amounts to 23.6%. The mechanical properties are directly related to rock strength behaviour, and data from mechanical properties provide high prediction accuracy. However, the physical properties of the enormous quantity of rock samples may sometimes be practicable, as AI models require extensive data. The mechanical properties were observed to be used as input parameters for AI models, provide better performance, reduce data complexity, and enhance models' interpretability in situations where there are limited datasets. Physical properties as input represent 17.3% of the research articles reviewed. The physical properties are easily determined and interpreted, and they are cost-effective in terms of resources and time compared to the mechanical testing of rock. They have a strong correlation in predicting UCS and E, but

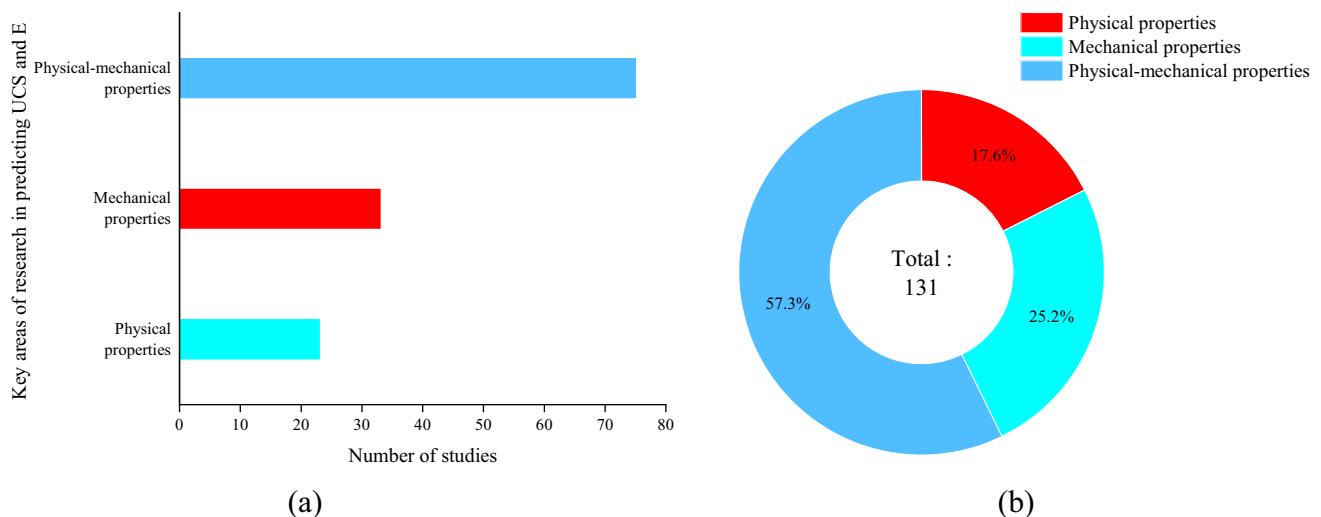


Fig. 5 Distribution of physical and mechanical properties and AI across subdomains in estimating UCS and E: **a** number of studies utilising AI, direct, and indirect rock properties in different key areas, **b** percentage distribution of rock properties employed as input parameters with AI models

some physical properties, including density and porosity, provide variability, which may influence the accurate prediction of UCS and E.

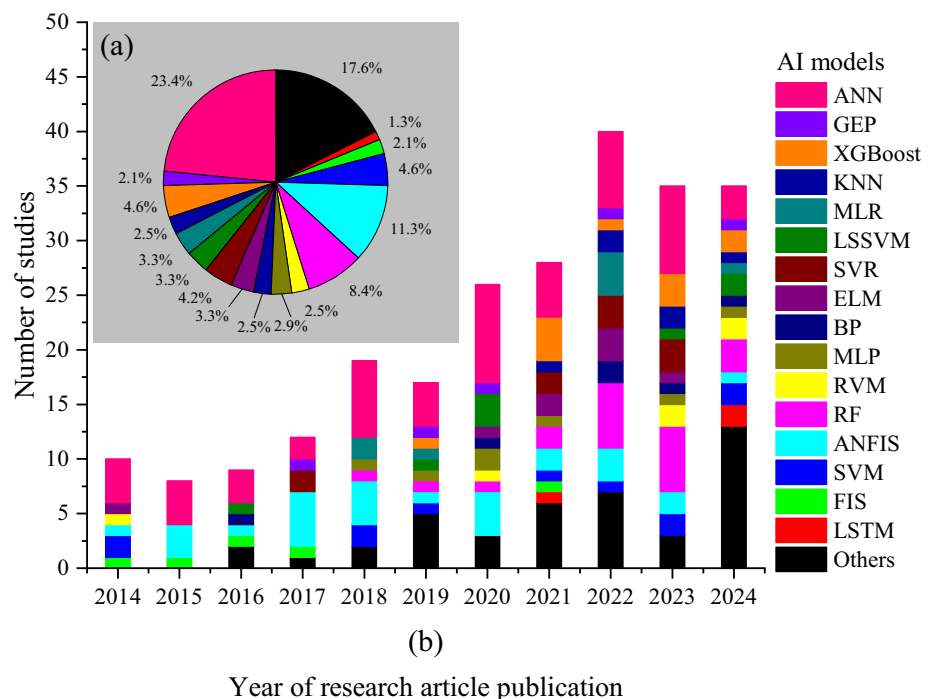
It should be noted that due to the large data requirement for AI models, most of the studies combined physical properties, mechanical properties, and a combination of physical–mechanical properties data from different sedimentary and igneous rocks from the same or different geological environments as input parameters for predicting UCS and E. The combination of different rock data from various sources was found more in the studies involved in the physical–mechanical properties compared to the physical and mechanical properties.

7.2 Statistical analysis of available AI algorithms

Several AI algorithms have been progressed over the past ten years, from 2014 to 2024. Figure 6 presents the distributions of various AI algorithms and a breakdown of all the AI techniques utilised in rock mechanics for estimating UCS and E. The most predominant are ANN-based algorithms (23.4%), ANFIS-based (11.3%), and RF (8.4%) algorithms employed in evaluating or predicting UCS and E with mechanical and physical properties, as displayed in Fig. 6(a). It can be observed from Fig. 6(b) that the use of ANN and ANFIS models has been adopted in predicting UCS and E from the 2014 fiscal year to date. The application of the RF model has also drawn attention from 2018 to date among researchers in the field of rock mechanics to estimate UCS, and E. Recent studies have also used models, such as XGBoost, RVM, and BP models due to their capability of predicting UCS and E. It is worth noting that in the stack column graph in Fig. 6(b), a single research article can utilise multiple AI models, resulting in some AI models used more than the number of papers reviewed in this study.

From the statistical point of view, ANN, ANFIS, SVM, RF, FIS, GEP, and XGBoost hold significant prominence in predicting UCS and E using rock mechanical and physical properties. Traditional ANN and ANN-based algorithms have been used especially due to their capability and robust adaptability in dealing with nonlinearities [7, 37, 39, 59, 60]. For instance, Ali et al. [54] predicted the UCS of amphibolite using microfabric properties, including grain size, quartz content, and shape factor. The study employed multivariate regression (MR), fuzzy inference system (FIS), and artificial neural network (ANN) and compared their predictive

Fig. 6 Breakdown of various AI techniques in the literature for predicting UCS and E: **a** distribution of AI algorithms adopted in the literature. **b** the research trends of various AI models utilised from 2014 to 2024



capabilities. After the evaluation of models according to performance indices like R^2 , RMSE, and VAF, all the models showed reasonable predictions. However, the ANN outperformed the other models with R^2 of 0.95, RMSE of 3.42, and VAF of 97.9%. Aghda et al. [42] also estimate the UCS and E of Asmari limestone with ANFIS and MR models utilising Brazilian tensile strength (BTS), porosity (n), and density (ρ) as inputs. The research uses 655 rock datasets, with 436 of the data used for model construction, and verifies the model with 219. The ANFIS after model construction and verification demonstrated high prediction accuracy compared to MR models, with VAF and RMSE values ranging from 90 to 95% and 6.86–8.94, respectively, demonstrating an excellent prediction of UCS and E compared to multiple regression methods.

Gowida et al. [33] explore the ANN, SVM, and ANFIS models to estimate UCS of rocks in real time during drilling operations using real-time drilling mechanical properties, such as standpipe pressure, penetration rate, rotary speed per minute, bit weight, mud pumping rate as models input parameters. Their studies demonstrated that the ANN model has the advantage over SVM and ANFIS models in predicting UCS in real time with high prediction accuracy with R^2 of 0.99 and AAPE of 3.5%. Using SVM and physical and mechanical properties of Maragheh limestone, Cemiloglu et al. [38] predicted the UCS of the rock with a high accuracy of 0.91, precision of 0.86, R^2 of 0.97, and a low error rate (MAE = 0.45, RMSE = 0.41 and MSE = 0.44) compared to the traditional fitting curves. The researchers suggested that the SVM model used is not suitable for large datasets and also not valid for probabilistic explanations for classification (SVM classifier). Matin et al. [35] explore the utilisation of the RF algorithm in predicting the UCS and E of travertine rock, focusing on selecting important variables used as the model input parameter (n , V_p , $I_{s(50)}$, and R_n). They concluded that the RF model can predict UCS and E with R^2 of 0.91 and 0.93, respectively. The V_p was found to be the most important variable in predicting UCS and E.

Heidari et al. [57] focus on developing predictive models to estimate the UCS of sedimentary rocks using multiple regression (MLR) and a Sugeno-type fuzzy inference system (FIS). The block punch index (BPI), V_p , $I_{s(50)}$, and R_n were used as model parameters. It was noted from the study that both FIS and MLR perform well in predicting UCS of the sedimentary rocks, with VAF of 90% for both models and RMSE of 10.80 and 12.82 for FIS and MLR, respectively. The GEP has been used successfully to estimate the UCS of sandstone with input indices such as n , BTS, and slake durability. The GEP was compared with linear multiple regression (LMR), and it was found to be superior to LMR with an R^2 of 0.965 [61]. Currently, an explainable artificial intelligence (XAI) such as XGBoost, AdaBoost, and CATBoost models and $I_{s(50)}$, R_n , porosity, and V_p were utilised to estimate the UCS of rock. The XGBoost among the three models identified as the best-performing model achieved an R^2 of 0.99 for training and 0.96 for testing, with $I_{s(50)}$ being a significant influence on prediction sandstone UCS [43].

In conclusion, AI technologies, with their exceptional capabilities and attributes, have a promising future in estimating the strength properties of rocks. However, AI algorithms have their peculiar limitations associated with them. For example, the ANN-based models have been identified as the best AI models for predicting the UCS and E of rocks due to their high accuracy. Despite their high performance, there are limitations, particularly data dependency, overfitting, and lack of interpretability, that need to be addressed to enhance real-time application and reliability in real-time prediction of UCS and E. Similar to ANN, all the tree-based models, such as XGBoost, AdaBoost, are data-dependent, complex in computation, require significant processing time, and do not generalise well to different types of rocks and geological conditions not represented in the training and testing data.

7.3 Physical and mechanical properties and AI models

Materials involved in rock mechanics and engineering are mainly rocks, which show a series of substantial physical–mechanical features, such as porosity, density, specific gravity, unconfined compressive strength (UCS), elastic modulus (E), Schmidt rebound hardness, point load, and block punch indexes. They collectively constitute rocks' physical and mechanical properties and serve as the core parameters for assessing rock strength and safety. The properties are usually determined from standard laboratory tests, including uniaxial compression, Brazilian

tensile, and point load tests [6, 54, 61, 62]. The UCS and E are essential for designing underground space, tunnel excavation, and dam sites [32, 41, 61, 63]. It is significant to note that rock materials are anisotropic and heterogeneous, and their physical and mechanical properties are influenced by temperature, stress path, and moisture or saturation. Determining the UCS and E directly from the unconfined compression tests is costly and laborious [33, 64]. Also, it is not easy to acquire quality rock cores from soft or highly fractured rocks [49]. Therefore, AI algorithms and indirect test results from the point load index ($I_{s(50)}$), Schmidt rebound hardness index (R_n), P-wave velocity (V_p), porosity (n), density (ρ), and block punch index (BPI) have been exploited to estimate UCS and E.

Figure 7 illustrates the research statistics on AI in predicting UCS and E with physical–mechanical properties. In recent years, significant increment in the research studies using AI models is to evaluate UCS and E with both the physical and mechanical properties of rocks, demonstrating the growing interest in the utilisation of AI in the field of rock mechanics and engineering. The primary use of AI models is ANN, ANFIS, and RF, representing 26.3%, 12.5%, and 10% of studies, respectively, from 2014 to 2024, underscoring the growing interest in the use of AI techniques for predicting rock strength.

Comprehensive strength and elastic modulus jointly establish the mechanical strength of rocks, which serves as part of the primary parameters for estimating rock strength, safety, and stability. Recently, AI models have demonstrated their efficiency in analysing and predicting the strength of different rocks, especially elastic modulus and compressive strength, considering different rock strength properties. The success of AI is due to its efficacy in processing and analysis of experimental datasets. Table 6 presents the usage of AI models in estimating the UCS and E of rock using physical–mechanical properties. In Table 6, the MAE, MAPE, RMSE, VAF, and R^2 are the AI model performance metrics to validate the accuracy of the various models used in the prediction of UCS and E. The AI model is efficient when MAE, MAPE, and RMSE are smaller and approaching 0. The increase in VAF and R^2 to 100 and 1, respectively, demonstrates the model's accuracy of the model data. For example, Jahed et al. [32] used the ANFIS model to forecast the UCS and E of granite with nondistractive parameters (i.e. ρ_d , V_p , Qtz, and Plg) as inputs. After comparing the model to MRA and ANN models, the ANFIS model demonstrated a prediction accuracy with a higher R^2 of 0.985, RMSE of 6.224, and VAF of 98.46% for UCS and R^2 of 0.99, RMSE of 3.503, and VAF of 98.97% for E.

Fig. 7 Breakdown of various AI techniques in the literature for forecasting UCS and E: **a** distribution of intelligence learning models adopted in the literature. **b** the research trends of various AI models utilised from 2014 to 2024

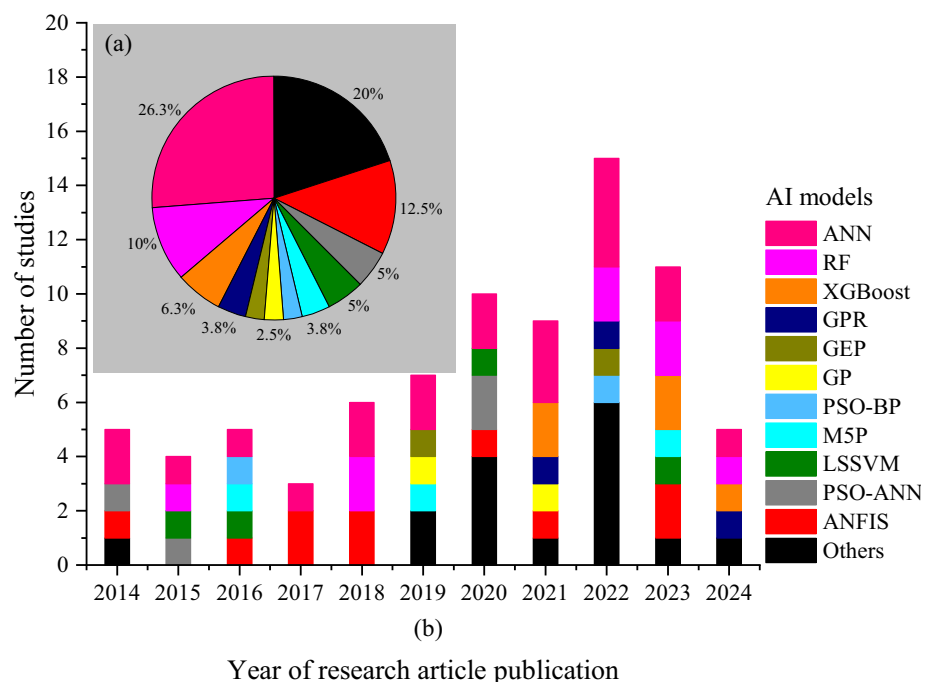


Table 6 AI algorithms for estimate of the UCS and E of rock applying physical–mechanical properties

Rock type	Sample size	Algorithms	Input parameter	Output	MSE	RMSE	MAE	VAF	R^2	References
Granite	45	ANFIS	ρ_d , V_p , Qtz, Plg	UCS, E	–	6.22, 3.50	–	98.50%, 98.97%	0.98, 0.99	Jahed et al. [32]
Limestone	105	ANN	n , ρ , V_p	UCS, E	–	7.10, 5.00	–	–	0.95, 0.76	Torabi-Kaveh et al. [63]
Limestone, granite	66	PSO-ANN	$I_{s(50)}$, R_n , V_p , ρ_d	UCS	0.004	–	–	97.24%	0.97	Momeni et al. [6]
Carbonate	54	M5P	R_n , V_p , SDI, ρ_d	UCS, E	–	10.47, 4.32	8.83, 3.40	70.17, 89.82	0.89, 0.84	Ghasemi et al. [71]
Carbonate	163	ANN	n , V_p , e , $I_{s(50)}$, ρ_d	UCS, E	–	3.80, 6.60	–	–	0.97, 0.96	Madhubabu et al. [72]
Sandstone	228	PSO-BP	ρ_d , V_p , $I_{s(50)}$, m_c	UCS	0.017–0.099	–	–	–	0.99	Mohamad et al. [73]
13-rock types	94	ANFIS	V_p , SDI, ρ	UCS	–	6.29	–	97.66%	0.98	Sharma et al. [74]
Basalt	80	ANFIS	V_p , ρ , n	UCS	–	0.51–3.23	–	76–98%	0.99	Singh et al. [75]
Limestone, sandstone, marlstone, conglomerate	196	ANN	ρ_d , V_p , n , A_b	UCS, E	–	0.094, 0.89	0.074, 0.071	–	0.93, 0.92	Abdi et al. [76]
Limestone	115	COA-ANN	v_d , ρ , V_p , n	UCS, E	1.93, 2.85	–	–	85.21%, 74.48%	0.98	Mokhtari et al. [69]
Travertine	93	ANN-committee	V_p , n , R_n , $I_{s(50)}$	UCS	–	3.98	3.23	–	0.89	Barzegar et al. [77]
Granite	71	GP	n , V_p , R_n , $I_{s(50)}$	UCS	–	0.0691	–	84.29%	0.84	Fang et al. [78]
Pyroclastic	50	GEP-I	ρ_d , V_p , n , ρ_s , $I_{s(50)}$	UCS	–	3.81	–	–	0.93	İnce et al. [79]
Marlstone	39	AdaBoost–NNE	V_p , ρ , V_s , BTS, $I_{s(50)}$	UCS, E	–	–	–	–	0.98, 0.92	Salehin et al. [80]
Sandstone	60	PSO-ANN	V_p , ρ , R_n	UCS	–	0.086	–	97.50%	0.97	Abdi et al. [40]
Basalt	56	ANN	ρ_d , V_p , R_n , $I_{s(50)}$, BTS	UCS	–	1.54	–	–	0.99	Barham et al. [81]
Dasitic, andesitic, tuff, basalt	47	MPMR	n , BPI	UCS	–	0.059	–	98.12%	0.96	Ceryan et al. [82]
Granite	88	GMDH	R_n , $I_{s(50)}$, n , V_p	E	–	7.19	–	95.07%	0.96	Armaghani et al. [83]
Travertine	32	ANN-PSO	ρ , V_p , BTS, V_s	UCS	–	1.37	–	85%	0.91	Ebdali et al. [62]
Shale, Iron pan	40	RVR-HS	$I_{s(50)}$, BTS, ρ_b , V_p	UCS	0.0031	–	–	–	0.99	Fattahi [84]
Shale	175	MLP-ANN, LSSVM-CSA	GR, DT, ρ_d , RT, n	UCS	–	0.174, 0.019	–	–	0.94–1.0	Miah et al. [70]
Travertine	29	ANN	V_p , n	UCS	0.054	–	–	–	0.66	Saldaña et al. [85]
Granite	182	BPNN	V_p , n	UCS	–	10.69	–	–	0.97	Armaghani et al. [60]

Table 6 (continued)

Rock type	Sample size	Algorithms	Input parameter	Output	MSE	RMSE	MAE	VAF	R^2	References
Granite, schist, sandstone, claystone, slate, marl, dolomite, travertine	170	GPR	$n, R_n, V_p, I_{s(50)}$	UCS	–	0.522	–	–	0.99	Mahmoodzadeh et al. [86]
Serpentinities	32	BPNN	$\beta, \gamma_d, \gamma_s, A_p, R_n, V_p, I_{s(50)}, V_s$	UCS	0.008	–	–	–	0.94	Moussas and Diamantis [87]
Travertine	30	XGBoost	$n, R_n, V_p, I_{s(50)}$	UCS, E	–	0.647, 0.037	–	–	0.99	Nasiri et al. [88]
Limestone, dolomite, onyx	30	ANN, ALPS-GP	$n, V_p, I_{s(50)}$	UCS	2.11, 1.64	2.50, 2.11	–	98.43%, 97.86%	0.98	Özdemir [89]
Sedimentary rocks	106	XGBoost	ρ_w, ρ_d, BTS	UCS	0.00	0.001	0.0054	–	0.99	Shahani et al. [90]
Igneous, sedimentary, metamorphic	60	RF	$R_n, n, BPI, I_{s(50)}, G_s$	UCS	–	–	8.86	–	0.94	Dadhich et al. [92]
Carbonate	94	SANN	γ, R_n	UCS	–	24.15	17.91	–	0.99	Hassan and Arman [93]
10 different rocks	147	ANN	n_e, ρ_d, V_p, UCS	UCS	–	3.87	–	94.09	0.94	Köken and Koca [94]
38 different rocks	93	ANFIS-GA	n, ρ, R_n	UCS, E	192.20	–	9.60	97.29	0.98	Rezaei and Asadizadeh [95]
Granite	71	ICA-ANN	$n, R_n, V_p, I_{s(50)}$	UCS	–	2.91	–	92.94	0.92	Armaghani et al. [9]

ρ_d dry density; V_p P-wave velocity; Qtz quartz; Plg plagioclase; n porosity; ρ density; $I_{s(50)}$ point load index; R_n Schmidt rebound hardness; SDI slake durability index; n porosity; D_p depth; e Poisson ratio; BPI block punch index; R^2 coefficient of determination; m_e moisture content; A_b water absorption; v_d dynamic Poisson ratio; ρ_s saturates density; V_s S-wave velocity; BTS Brazilian tensile strength; V_s S-wave velocity; ρ_b bulk density; GR gamma ray; DT sonic tensile time; RT true resistivity; β serpentinisation percentage; γ_d dry unit weight; γ_s saturated unit weight; G_s specific gravity; γ dry unit weight; T temperature; E_d dynamic Young's modulus; n_e effective porosity; UCS unconfined compressive strength

In another study, Köken [34] employed CART, ANN, GEP, MARS, and ANFIS models with ρ_d , n , and V_p as input parameters to estimate the UCS of pyroclastic rock. The researchers concluded that the models performed better with R^2 ranging from 0.82 to 0.88, demonstrating the model's reliability in predicting the UCS of pyroclastic rocks. In addition, they suggested that the models used in the study were limited to pyroclastic rocks only. Heidarian et al. [65] also applied the ANFIS model to predict the UCS of rock, considering the porosity (n), depth (D_p), and P-wave velocity (V_p). After training, testing, and validation of 655 samples, the ANFIS model proved to be effective in the prediction of the UCS. Asheghi et al. [66] pioneered a hybrid AI algorithm, a generalised feedforward network (GFFN) with ICA, for an accurate estimate of the UCS of quarry rocks. They concluded that the ICA-GFFN model shows better performance in predicting UCS, indicating the model's capability after comparing it with standard AI models like GFFN, RBF, and MLP.

Additionally, among the input parameters such as n , V_p , $I_{s(50)}$, and water absorption, the V_p was identified as the significant input factor in predicting UCS according to sensitivity analysis. Li and Tan [67] employed the LSSVM

model to assess the UCS of different rocks. They researched 40 samples and found LSSVM effective in predicting UCS with n , SDI, V_p , and $I_{s(50)}$ as model inputs. Khan et al. [68] exploited RFR, ANN, KNN, and MLR models to forecast UCS and E, using T , V_p , n , ρ , and E_d as input parameters. After analysing 60 samples of calcite and dolomite, the RFR model emerged as a suitable model for calculating UCS and E with R^2 of 0.97. However, the proposed model is suitable for estimating the UCS and E of high-temperature treated rock. Other studies have explored the ANN model and observed that it performs more advantageously than the multivariable regression model. The hybrid optimisation of ANN and other algorithms, including COA-ANN, PSO-ANN, MLP-ANN, PSO-LSSVM, and ICA-ANN, have also been reported in the studies of [40, 48, 69, 70].

The $I_{s(50)}$, R_n , density (ρ), specific gravity (G_s), and water absorption (A_b) for predicting UCS and E. For example, this study summarises some of the AI algorithms and physical–mechanical properties in Table 7, including model accuracy measurement metrics and limitations. Asare et al. [96] estimated UCS with the AutoNN-MARS model, where the model integrates the AutoNN to extract feature and dimensionality reduction with MARS for regression analysis using R_n , $I_{s(50)}$, ρ_b , and BTS as input variables. They compared the model with four different existing AI models: RF, BPNN, and GRNN. The researchers concluded that the AutoNN-MARS model outperforms all the other models with high precision accuracy, demonstrating an r -value of 0.99, RMSE of 0.39, and P_i of 0.31. Koopialipour et al. [97] developed a stack-tree-KNN-RF-MLP model using different MLs to estimate E. The model had excellent accuracy results with R^2 of 0.83, RMSE of 15, MSE of 227, and MAE of 12. However, the model can only be used with the input parameter range employed in their research.

Gene expression programme (GEP) generates an interpretable structural algorithm and is applicable to predict rock strength, excelling in handling uncertainties and nonlinearities of rock data. For instance, Xue [98] developed GEP based on 44 rock datasets and evaluated it with four traditional regression models, including the ANN model. The comparison results of the study indicated that GEP performance was better than that of the other models, with the lowest RMSE of 10.22 and R^2 of 0.98 in predicting UCS. The results demonstrate the GEP model's potential in predicting rock strength by utilising BPI, R_n , $I_{s(50)}$, and V_p as model input. Malkawi et al. [30] predicted UCS of travertine rock using rock indices, including R_n , Leeb rebound hardness, $I_{s(50)}$, absorption, pulse velocity, porosity, and density as input parameters. They used 61 data with ML models such as M5T, ANN, and KNN. Among the models, the M5T outperformed ANN and KNN in predicting UCS with MAPE of 0.034, RMSE of 0.045, and Spearman's rho of 0.96. In order to estimate the UCS of rock, Lei et al. [99] use six hybrid AI models integrated with a back-propagation (BP) neural network and five input parameters (E , n , R_n , $I_{s(50)}$, and V_p). The study reported that among the six models, the firefly algorithm-BP (FA-BP) hybrid model demonstrated the best prediction accuracy of R^2 of 0.985, RMSE of 4.88, MAPE of 0.063, and 10-index of about 0.967. Additionally, the sensitivity analysis of input parameters revealed that $I_{s(50)}$ is the most important parameter for estimating UCS.

Xu et al. [100] developed an AI model based on a sparrow search algorithm (SSA) optimised with XGBoost. The SSA-XGBoost model integrates SSA to optimise the hyperparameters of the XGBoost model to improve the model predictive accuracy using R_n , $I_{s(50)}$, and V_p and n as model inputs. Their model was evaluated against six different AI models. According to the evaluation measurement performance metrics, the SSA-XGBoost outperformed the other models with R^2 of 0.96, RMSE of 0.36, MAPE of 11.89%, MAE of 7.06, RMSE of 9.42, and VAF of 95.97%, indicating the high performance of the model in predicting UCS. Despite the model prediction performance, the model is limited to careful hyperparameter tuning, potential generalisation problems with different data and complexity of model computation. Wu et al. [101] proposed optimised kernel ridge regression (KRR) and Gaussian process regression (GPR) models based on the grid-search method for estimating UCS and E. The KRR demonstrates exemplary performance in predicting UCS with an average error of 4.9%, and the GPR model demonstrates better prediction of E with a relative average error of 1.10% using meso-mechanical properties as model input parameters.

By using physical–mechanical properties from direct and indirect test results as input parameters, including R_n , dry density, block punch index, BTI, water absorption, porosity, and $I_{s(50)}$, AI algorithms can capture a complex nonlinear relationship and accurately estimate E and UCS of rocks. The estimation outcomes help rock engineers

Table 7 AI algorithms for the forecast of the UCS and E of rock utilising mechanical properties

Model	Samples	Inputs	Output	R^2	MSE	RMSE	MAE/ MAPE	VAF (%)	Limitation	References
Stack- tree- KNN- RF- MLP	92	$n, I_{s(50)}, R_n, V_p$	E	0.83	277	15	12	–	The model can only be used with the input parameter range employed in this research	Koopialipoor et al. [97]
ANN- ICA	367	n_e, R_n, V_p	UCS	0.98	–	11.94	0.1196	95.10	The model is limited to the input parameters used, and any value outside the minimum and maximum values cannot provide a reliable prediction	Le et al. [102]
KELM- GWO	271	$n, R_n, V_p, I_{s(50)}$	UCS	0.92	–	14.73	11.43	91.52	The prediction performance is limited by parameter diversity	Li et al. [103]
TSO-RF	226	$n, R_n, V_p, I_{s(50)}, \rho_d$	UCS	0.97	–	6.60	–	–	The model is limited to specific rock types adopted by the researchers and may not be considered a general model for other rocks	Li et al. [104]
SVR	115	V_p, n, ρ, μ_d	UCS, E	0.70, 0.76	22.15, 0.29	–	–	–	The model is limited to carbonate rocks, and it should be used with quadratic kernel and cubic kernel functions	Mokhtari [105]
WOA- ELM	734	$R_n, V_p, I_{s(50)}$	UCS	0.86	17.61	4.20	–	91	The model has local minima trapping issues	Qiu et al. [106]
ANN- LM	274	n_e, R_n, V_p	UCS	0.96	–	14.83	0.11	94.30	Potential variations of input parameters could influence the model. For example, V_p is limited between 1000 to 3000 m/s	Skentou et al. [28]
PSO- SVR	131	$L, D, \text{grain size, strain rate, } \rho_b, \text{SCS, } V_p$	UCS	0.99	–	–	4.95, 4.72	–	Increasing input parameters like bulk density and strain rate increases UCS	Yang et al. [47]
GA–SEL	161	$BPI, R_n, I_{s(50)}, V_p$	UCS	0.92	–	13.45	–	91.01%	Limited dataset and does not cover all types of rock	Zhang et al. [107]
fB-GPR	71	$n, R_n, V_p, I_{s(50)}$	UCS	0.90	–	13.24	–	–	The model has a large magnitude of measurement errors	Zhao et al. [108]
XGBoost	1771	V_p, n_e, ρ	UCS	0.91	–	439	17.22	–	The model is limited to carbonate rock. Parameters like grain size and mineral composition were not considered	Abdelhedi et al. [64]
SVM	50	$RAI, AIV, LAAV, R_n, I_{s(50)}$	UCS	0.90	–	1.16	0.76	–	Limited rock samples restrict the ability of the model	Afolagboye et al. [109]
ANFIS	45	$I_{s(50)}, BTS, BPI$	UCS	0.99	–	0.36	0.28	99.92	The model is valid for limestone, travertine, sandstone and conglomerate only	Khajevand [110]
LSO-RF	386	$V_p, n, I_{s(50)}, R_n$	UCS	0.92	–	13.75	–	92	The initialisation of a random population tends to trap it into local minima optimisation	Li et al. [111]

Table 7 (continued)

Model	Samples	Inputs	Output	R^2	MSE	RMSE	MAE/ MAPE	VAF (%)	Limitation	References
SSA-RF	126	V_p , n , $I_{s(50)}$, R_n	UCS	0.96	–	9.42	7.06, 11.89		Low-quality data size	Wang et al. [112]

n_e effective porosity; R_n Schmidt hammer rebound number; V_p P-wave velocity; n porosity; μ_d dynamic Poisson ratio; BPI block punch index; R^2 coefficient of determination; SCS static compressive strength; Em elastic modulus; ρ_d dry density; $I_{s(50)}$ point load index; ρ density; L length; D diameter; SCS static compressive strength; V_p P-wave velocity; n_e effective porosity; ρ density; RAI rock aggregate index; AIV aggregate impact value; $LAIV$ Loss Angeles aggregate value; BTS Brazilian tensile test; ρ_d dry density; A_b water absorption; r correlation coefficient; P_i performance index

understand and determine the rock UCS and E at numerous conditions and augment model and testing methods, therefore enhancing efficacy. In addition, the utilisation of AI algorithms is not restricted to experimental data handling in predicting rock mechanical properties. The models can be extended to predict and monitor real-time field conditions. The application of AI models in rock engineering ensures that the predictions of rock strength properties are consistent and accurate with different complex geoenvironmental and loading conditions. However, some of the AI models developed in the literature are case-specific and may not be applicable to predict UCS and E of different rocks with different geological conditions. Therefore, in selecting an AI model from the literature, careful analysis must be performed to determine the model's capability and reliability in performing the task.

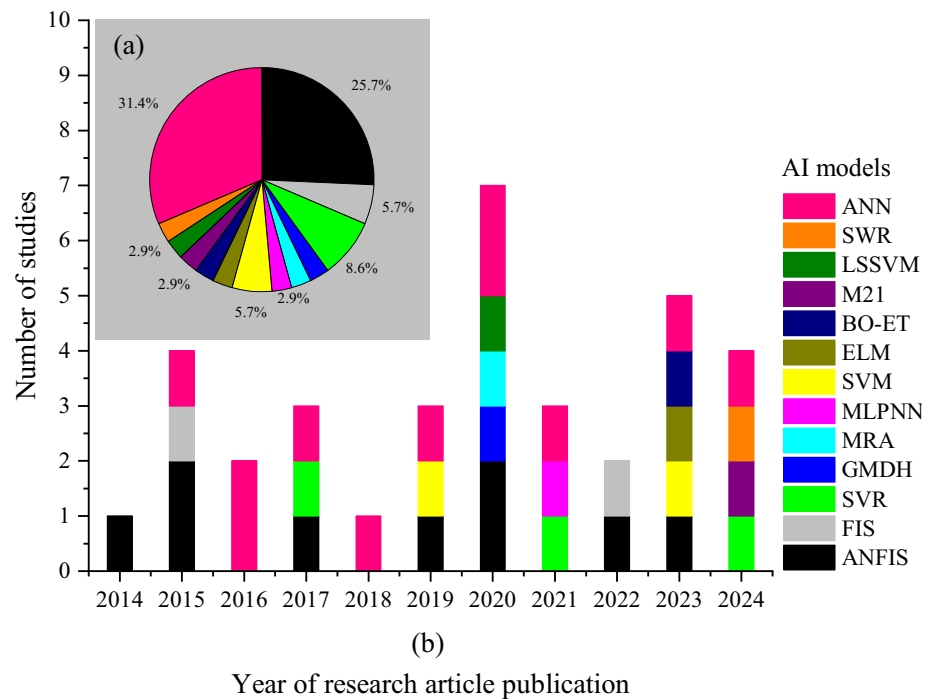
7.4 Mechanical properties and AI models

Rock exhibits a range of critical rock mechanical properties such as UCS, e , TS, $I_{s(50)}$, R_n , BTI, V_p , and E. These mechanical properties not only influence the stability of rock engineering structures but also play a critical role in the design and construction of geomechanics projects. These mechanical properties have been widely used to predict UCS and E [35, 91, 113–115]. Figure 8 shows statistical data on research involving AI models in predicting UCS and E with mechanical properties of rocks from 2014 to 2024. As depicted in the figure, common AI models include ANN (31.4%), ANFIS (25.7%), and SVR (8.6%).

AI models have shown the potential to predict UCS and E with rock mechanical properties. Kaithola et al. [113] exploited the ANFIS model and linear regression with V_p to estimate the UCS and other rock properties from the Himalayan. The ANFIS model outperforms the traditional linear regression models in predicting UCS, indicating higher accuracy and reliability. Mohammadi et al. [116] utilised hybrid ANFIS-PSO and ANFIS-GA to predict UCS and E. They noticed a successful prediction of UCS and E of dolomite and carbonate rocks, but the model was less accurate in predicting the Poisson ratio and internal friction angle. The ANFIS-PSO model demonstrates high prediction accuracy with a low RMSE of 1.14 and R^2 of 0.90 than the ANFIS-GA model, considering weight on the drill bit (WOB), drill string rotation speed (RPM), rate of penetration (ROP), and torque (Trq) as model input parameters. A similar study by Davoodi et al. [117] explored the least square support vector machine (LSSVM) and multi-layer extreme learning machine (MELM) algorithm amalgam with COA, PSO, and GA to forecast UCS using depth (L), ROP, RPM, WOB, Trq, bit wear rate (CT), and flow rate (FLR) as model input. After the construction, training and testing of 3042 datasets from two wells, the MELM-COA model achieved high prediction accuracy in estimated UCS with R^2 of 0.99 and RMSE of 4.69. The L , ROP, RPM, WOB, and Trq were found to influence the prediction of UCS significantly.

Gamal et al. [118] developed AI models RF based on Principal Component Analysis (PCA) and Function Network (FN). The drilling data include WOB, drill string rotating speed (RS), drilling torque (T), stand-pipe pressure (SPP), mud pumping rate (Q), and ROP as model input parameters to predict real-time UCS of rock. The RF-PCA model surpassed the RF-FN model, achieving an R^2 of 0.99 and an average absolute percentage error (AAPE) of 4.3%. In comparison, the FN model had an R^2 of 0.97 and an AAPE of 8.5%. Validation results were

Fig. 8 Breakdown of AI algorithms in forecasting UCS and E with physical–mechanical properties: **a** distribution of different algorithms employed in estimating rock UCS and E, and **b** trends and number of studies involving different AI models in using physical–mechanical properties from 2014 to 2024



consistent, with the RF model demonstrating higher accuracy. Siddig et al. [119] also used three AI models: ANFIS, RF, and FN with WOB, drill pipe torque, and ROP to estimate static elastic modulus (E_s). The models showed high accuracy, with R^2 ranging from 0.92 to 0.99 and AAPE below 13%. The results indicate that using drilling data and AI techniques to predict E_s is promising. This approach is a cost-effective and efficient alternative to traditional methods, providing continuous profiles of rock mechanical properties crucial for applications like fracturing design and wellbore stability.

In addition, Hiba et al. [120] predicted UCS with ANN using input parameters similar to [118, 119]. The ANN model demonstrated high accuracy, with AAPE of 0.28% during training and 0.59% during testing, effectively predicting UCS in real time. This AI-based approach provides a fast, reliable, and continuous method for predicting rock strength parameters, offering significant advantages over traditional laboratory-based methods.

Daniel et al. [121] developed models using GBDT, XGBoost, LightGBM, RF, and ET hybridise with Bayesian Optimisation (OB) for predicting the UCS of metamorphic, magnetic, and sedimentary rocks, utilising input parameters such as R_n , V_p , and $I_{s(50)}$. Notably, after analysing 729 datasets, the BO-ET optimisation algorithm outperformed the other five models with R^2 of 0.99, VAF of 99%, MAE of 3.97, and RMSE of 4.82. Also, a graphic user interface (GUI) system was developed based on the BO-ET model and validated in tunnel boring machine excavated tunnel. Khatti and Grover [122] introduced the MD21 model, including a PSO-optimised relevance vector machine (RVM) model implemented by Laplacian + polynomial for predicting UCS with nondestructive mechanical properties.

After training and testing 734 rock datasets, the MD21 model demonstrates an optimal performance in validation and testing with R^2 of 0.998 and RMSE of 1.93 compared to the traditional models. The MD21 model offers a highly accurate and reliable method for estimating UCS. It provides valuable insights for geotechnical engineering applications. Despite the performance of the M21 model, there exists a limitation: using a hybrid learning model to determine the optimal structure, which will be utilised in numerous analyses. Table 8 details the application of specific AI models in assessing the UCS and E of rock with mechanical properties. From two ML models, step-wise regression (SWR) and simple linear regression (SLR) for predicting UCS of Charnockite rock, Kochukrishnan et al. [115] used indirect test results (R_n , BTS, V_p , and $I_{s(50)}$) as input parameters and SWR

outperformed the SLR achieved a low MAE of 1.71 and R^2 of 0.99. In the ranking of the significance of the input parameters, the V_p has a significant influence on estimating the UCS of the Charnockite rock.

ANN, ANFIS, and other algorithms have substantial advantages in predicting UCS and E. Research by Jan et al. [133] demonstrated that the UCS of marble rocks could be predicted with AI models: ANN, RF, XGBoost, SVM, Elastic net (EN), Ridge, and Lasso model using mechanical properties as input parameters. The result indicates that the ANN model had the highest prediction accuracy with R^2 of 0.99, RMSE of 0.26, MSE of 0.069, and MAE of 0.164 compared to other models. However, the XGBoost model also showed a close performance in predicting UCS as the ANN. The models were trained and tested using limited rock data and three features, which might not be able to capture significant information in predicting UCS accurately. Also, since the study used only marble rock, the models may be fine-tuning when applying it to other rock types.

In predicting E, the RMSE of the ANFIS was 0.0025 for coal rock, 0.127 for ANN with sandstone, and 0.82 for SVR with limestone [27, 124, 127]. These results indicated that ANN and ANFIS offer high predictive accuracy and efficiency in dealing with large datasets and complex nonlinear relationships in real-time rock data. The ANN and ANFIS excel at capturing the UCS and E of rocks and their influencing factors. SVR also performs better with small and linear data, but it faces challenges with nonlinear and enormous datasets, which makes SVR less tractable than ANN algorithms. However, their estimated error can sometimes be less than those of other AI algorithms. The SVR model has shown its effectiveness and suitability for predicting UCS due to a higher computational advantage [124]. Overall, ANN and ANFIS outperform the SVR in efficiency, accuracy, and adaptability. Nonetheless, future studies should emphasise the strength of these algorithms in developing efficient and hybrid algorithms.

Table 8 AI algorithms for evaluation of the UCS and E of rock employing mechanical properties

Rock type	Sample size	Algorithms	Input parameter	Output	R^2	RMSE	References
Limestone, schist, quartzite, slate	597	ANFIS	V_p	UCS	0.91–0.97	–	Kainthola et al. [113]
Granite, schist, sandstone	44	FIS, ANN, ANFIS	BPI, $I_{s(50)}$, R_n , V_p	UCS	0.92–0.98	9.54–16.90	Mishra et al. [123]
Sandstone	96	ANN	$I_{s(50)}$, R_n , V_p	E	0.81	0.127	Bejarbaneh et al. [27]
Sandstone	108	ICA-ANN	R_n , V_p , $I_{s(50)}$	UCS	0.95	3.25	Armaghani et al. [15]
Limestone	482	SVR	E_d , v_d	UCS, E	0.92, 0.93	1.31, 0.82	Aboutaleb et al. [124]
Basalt, metabasalt	47	ANFIS-SCM	R_n	UCS	0.93	0.198	Fattahi [41]
Calcareous mudstones	80	ANFIS	BPI, $I_{s(50)}$, CPI	UCS, E	0.96, 0.79	–	Mahdiabadi and Khanlari [125]
Migmatite	120	ANN	CPI, BPI, BTS, $I_{s(50)}$, V_p	UCS, E	0.90, 0.93	–	Saedi et al. [126]
Coal	69	ANFIS	CS, TS, SS, V_p	E	0.98	0.0025	Roy et al. [127]
Granite	96	SFS-ANFIS	R_n , V_p , $I_{s(50)}$	UCS	0.98	6.65	Jing et al. [128]
Granite	100	GMDH	R_n , V_p , ρ_d	UCS	0.86	0.089	Li et al. [29]
Different rocks	93	MRA	R_n , BTS, SH, $I_{s(50)}$, V_p	UCS	0.91	–	Teymen et al. [129]
Granite, schist, sandstone	44	GA-LSSVM	BPI, R_n , V_p	UCS	0.98	6.68	Xue and Wei [130]
Basalt, dolostone, limestone	30	MLPNN	SH, BTS, V_p	UCS	0.99	1.34	Gül et al. [131]
12 different rocks	37	ANN	V_p	UCS	–	–	Rahman and Sarkar [132]

V_p P-wave velocity; BPI block punch index; $I_{s(50)}$ point load index; R_n Schmidt rebound hardness; R^2 coefficient of determination; E_d dynamic Young's modulus; UCS unconfined compressive strength; CPI cylinder punch index; v_d dynamic Poisson ratio; CS compressive strength; TS tensile strength; SS shear strength; BTS Brazilian tensile strength; SH shore hardness; E elastic modulus; SCM subtractive clustering method

In recent decades, AI algorithms such as ANN, RF, and SVM have demonstrated remarkable success in predicting UCS and E of different rock types. Several studies have shown that the combination of these algorithms with optimisation algorithms offers significant benefits in terms of forecasting UCS and E of rocks, for example, the optimisation of ANN with ICA and marine predator algorithms (MPA). In addition, Jing et al. [128] combined hybrid models such as particle swarm optimisation (PSO), subtractive clustering method (SCM), genetic algorithm (GA), stochastic fractal search (SFS), and differential evolution (DE) with ANFIS for predicting UCS of rock using V_p , R_n , and $I_{s(50)}$. They concluded that the SFS-ANFIS model had high accuracy in predicting UCS with R^2 of 0.98, MAE of 4.44, and MAPE of 4.82 compared with the other hybrid models.

Furthermore, for the UCS forecast, the ANN algorithms surpassed the performance of MVRA, and fusion optimisation models, with ICA-ANN and PSO-ANN, further boosted the algorithm's performance [32, 73, 134]. Table 9 shows a comparison of hybrid models developed in the literature. From the table, it is clear that the hybrid models provide an excellent tool for predicting the UCS and E of rocks, but the generalisation of these models is still at the primary stage, because the same rock with different model input parameters may present different model outcomes (i.e. UCS and E) when using the same AI model.

In summary, AI techniques have demonstrated significant potential in predicting UCS and E with rock mechanical properties. The AI algorithms for predicting UCS and E include extensive experimental data, like point load index, unconfined compressive, Brazilian tensile, block punch, Schmidt hammer rebound, and P-wave and S-wave velocity tests. The datasets from these tests, AI models can be constructed, trained, and tested to capture nonlinear relations, predict real-time UCS and E, and provide accurate prediction results. It should be noted that most of the AI models discussed cannot be generalised for predicting UCS and E due to different model input parameters used, which may require a significant modification. It has also been found that limited datasets can impact the AI model's performance in predicting UCS and E. As presented in Table 8, the AI model shows excellent performance in predicting UCS and E. However, caution must be considered when selecting an AI model to perform any predictions in terms of rock strength. Most of the studies combined datasets from different rock types, which makes it challenging to interpret and identify the particular rock type.

Table 9 Current AI hybrid models in literature for predicting UCS and E of rocks

AI hybrid model	Output	Performance (R^2)	References
ICA-GFNN	UCS	0.98	Asheghi et al. [66]
PSO-ANFIS	UCS	0.96	Jin et al. [128]
GA-ANFIS		0.94	
DE-ANFIS		0.95	
SFS-ANFIS		0.98	
ANFIS-SCM	UCS	0.93	Fattahi [41]
SVR-ABC		0.97	
ICA-ANN	UCS	0.95	Armaghani et al. [15]
XGBoost-FA	UCS, E	0.99, 0.97	Cao et al. [37]
ANN-PSO	UCS	0.97	Momeni et al. [6]
ANN-PSO	UCS	0.97	Mohamad et al. [39]
ANN-PSO	UCS	0.99	Sabri et al. [49]
ANN-TLBO		0.93	
WOA-ELM	UCS	0.86	Qiu et al. [106]
MELM-COA	UCS	0.99	Davoodi et al. [117]
GA-SEL	UCS	0.92	Zhang et al. [107]
COA-ANN	UCS	0.97	Mokhtari and Behnai [69]
PSO-LSSVM	UCS	0.95–0.99	Wen et al. [48]

7.5 Physical properties and AI models

The physical properties of rocks significantly influence the engineering characteristics of rocks, including bulk density [135], quartz [37, 136], CT images [44], discontinuity density [137], and porosity [82], among others. As presented in Fig. 9, AI techniques are widely used in estimating the UCS and E of rock with physical properties. However, not many studies have focused only on the physical properties of rocks in determining UCS and E, as in Fig. 9(a). The percentage ratios of the AI algorithms used with physical properties in predicting UCS and E are shown in Fig. 9(b). It is evident from Fig. 9(b) that AI models: ANN, RF, ANFIS, and SVM models have seen growing interest in predicting UCS and E utilising rock's physical properties as the model input parameters.

Sun et al. [36] proposed an innovative technique for reconstructing 2D fracture characteristics and mineral distribution of mudstone from X-ray CT images processed with the LOG algorithm. The study introduced the CNN algorithm with stochastic pooling technique and applied the model to LOG images to predict the UCS of the rock. The study revealed that the CNN model can predict the UCS of the mudstone with CT slices with a mean error of 2.19%. However, the proposed method used depends on significant data for high-resolution X-ray scanning. Overall, this proposed model supports the performance of UCS prediction.

Wu et al. [138] also employed X-ray CT images of limestone with porosity, dry density, P-wave velocity, and other derived parameters as input to predict UCS. They utilised SVM, RF, and BPNN-based models embedded with genetic algorithms (GA) to optimise the input parameters and effectively predict the UCS. Among other models, the BPNN-base algorithm performed excellently with R^2 of 0.92, MAPE of 14.81, RMSE of 8.68, and MAE of 6.85. Through AI methods, researchers can automatically acquire and analyse rock strength with mineral content [24, 139], rock quality designation [137], high–low-frequency ratio, and amplitude attenuation coefficient [140]. These are indicators forming the fundamentals for analysing UCS and E with some physical properties. The AI technology also enables the reconstruction of 2D images and provides a quick and accurate way of acquiring parameters. The technology provides solid grounds for research prospects and the use of AI algorithms in rock engineering.

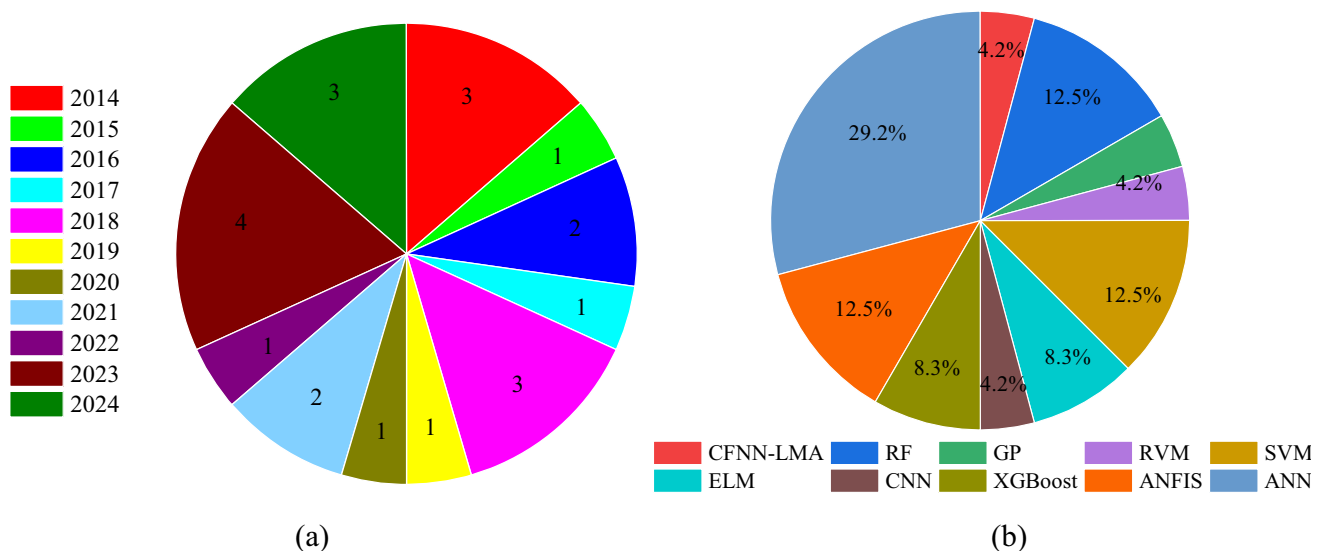


Fig. 9 Breakdown of AI algorithms in predicting UCS and E with mechanical properties: **a** distribution of various algorithms employed in estimating rock UCS and E and **b** trends and number of studies involving various AI models in using mechanical properties from 2014 to 2024

Elkakatny et al. [135] conducted a pioneering study using learning models such as ANFIS, SVM, and ANN to predict E using bulk density (ρ_b), compression time (C_t), and shear time (S_t) as input parameters. Among these models, the ANN algorithm shows the best prediction accuracy of R^2 of 0.96 and AAPE of 6.2%. In a similar study, Wang et al. [141] applied dry density (ρ_d), quartz, mica, and plagioclase as input, and the XGBT model attained an R -value of 0.99 and MAE of 0.11 in predicting UCS. Hasanipanah et al. [137] successfully optimised CFNN with LMA to predict the deformation modulus of rock using depth, RQD, discontinuity orientation adjustment, groundwater conditions, and discontinuity density as model input.

Feldspar, fragments, and quartz minerals were employed as model input parameters by Fang et al. [136] to predict UCS and E with the ANFIS algorithm ($R^2 = 0.99$). The SVM provides excellent results in predicting compressive strength. Utilising parameters including amplitude attenuation coefficient and high–low-frequency ratio, the R^2 reaches 0.99, and MAE and RMSE were 0.092 and 0.10, respectively [140]. Hybrid intelligence systems, such as artificial neural networks with particle size optimisation (ANN-PSO) and teaching learning-based optimisation (ANN-TLBO) using unit weight and petrographic analysis, result in higher R^2 and low Theil's inequality coefficient (TIC) [49]. Table 10 summarises some of the studies involving physical properties.

In summary, AI techniques demonstrate promising potential in predicting UCS and E of various rocks with physical properties, including density, mineral contents, X-ray CT, and petrographic images as input parameters. The input data can be acquired from CT scans and other laboratory tests, and by training these data, the AI algorithms can capture both complex nonlinear and linear relationships to predict the UCS and deformation characteristics of rocks. AI technology can provide accurate predictions for rock discontinuities (fragments or fractures), water saturation, P-durability index, and neutron porosity. The application of AI extends to real-time monitoring, ensuring that predictions of rock strength with physical properties are dependable considering different complex geological terrains and stress conditions.

Table 10 Summary of some studies on AI algorithms for evaluating the UCS and E with physical properties

Rock type	Sample size	Algorithms	Input parameter	Output	R^2	References
Carbonate	120	ANFIS	ρ_b , C_t , S_t	UCS, E	0.94, 0.88	Tariq et al. [142]
Granite	45	XGBoost-FA	V_p , Qtz, Kpr, Plg, Chl, Mica, ρ_d	UCS, E	0.99, 0.97	Cao et al. [37]
Carbonate	2600	RF	ρ_b , GR, DTC, DTS, n_p	UCS	0.99	Ibrahim et al. [143]
Shale	5000	GP	n , W_s , ρ_b	UCS	0.96	Koolivand-Salooki et al. [144]
Sandstone	64	ANFIS	Qtz, Fr, Fg, V_p	UCS, E	0.99	Fang et al. [125]
Granite	45	XGBT	ρ_d , Qtz, V_p , mica, Plg	UCS	0.99	Wang et al. [141]
Dasitic, andesitic, turffs and basalt	10	RVM	n , PDI	UCS	0.85	Ceryan [82]
Carbonate	54	ELM	MC, ρ_s , γ_d , n_t , n_e	UCS	0.79	Liu et al. [24]
Granite	54	ANN-PSO	γ_d , PA, MC	UCS	0.99	Sabri et al. [49]
Mudstone	15	CNN	150 CT images	UCS	CT-value % range from 0.94 to 0.96	Sun et al. [44]
Granite	40	SSA-XGBoost	V_p	UCS	0.94	Xie et al. [145]

ρ_b bulk density; C_t compression time; S_t shear time; Qtz quartz; Kpr alkali feldspar; Plg plagioclase; Chl chlorite; ρ_d dry density; V_p P-wave velocity; GR gamma ray; DTC sonic time compression; DTS shear time; n_p neutron porosity; n porosity; W_s water saturation; Fr feldspar; Fg fragments; PDI P-durability index; MC mineral content; ρ_s specific density; γ_d dry unit weight; n_e effective porosity; n_t total porosity

8 Limitations and challenges

The primary challenge of AI algorithms is the reliability of large datasets, which is the critical component for the training and testing of AI algorithms. Data quality is crucial for the efficiency and strength of AI algorithms, as it significantly impacts their performance. In addition, the simplification of AI algorithms necessitates evaluation and authentication. Moreover, the interpretation and explanation of AI algorithms are essential. Comprehending how to explain AI model expectations is significant for extending confidence and warranting realistic application in rock engineering. The challenge lies in ensuring the precision and reliability of AI algorithms in reasonable applications, thereby positively impacting rock mechanics and engineering.

AI algorithms significantly enhanced the computation capabilities of processing data and recognised forms. AI algorithms face considerable difficulties in accommodating complex rock mechanics scenarios and varying geological terrains. With data scarcity, AI algorithms sometimes face challenges in providing accurate and credible evaluations of rock strength. This is because the AI algorithms mainly depend on data-driven methods, not integrating physical concepts, which sometimes weakens the algorithm's interpretability and reliability in predicting complex rock strength characteristics and other geological conditions. These AI algorithms are often used to train data on specific rocks from different complexities of real-world geological formation environments that may not adequately epitomise the diversity or to generalise and stipulate accurate forecasting with diverse rock mass characteristics that can be co-operated.

9 Future research directions

AI algorithms offer significant opportunities due to their exceptional analytical capabilities to assist in evaluating complex and nonlinear relationships to rock engineering, hence offering crucial evidence that standard modelling techniques sometimes miscue. AI technologies ease the integration and examination of diverse data from the mechanical and physical properties of rocks, ranging from diverse laboratory tests to the promotion of an all-inclusive, multi-scale elucidation of rock strength characteristics, which allows for the optimisation of test designs and methods. The future research directions may include:

1. Many studies have combined physical–mechanical properties in predicting UCS and E as input parameters. On the other hand, few studies focused on the use of physical and mechanical properties with AI algorithms in estimating UCS and E. It is recommended that more studies are conducted on physical and mechanical properties only, especially physical properties since they are easy to determine. Additionally, the influence of individual parameters of both physical and mechanical properties is used as model input in predicting rock UCS, and E should be investigated to supplement the literature.
2. By combining adaptive learning strategies and the physical concept of rock behaviour into computational methods by integrating deep learning and fundamentals of rock mechanics principles, deep learning AI algorithms can obey physical rules and theories, such as Drucker–Prager model, Bieniawski's RMR, and Hoek–Brown's criterion as well as incorporation of stress–strain relationship to create synthetic features (e.g. mineralogy, texture, porosity, hardness, and lithology) that estimate UCS based on other rock properties for training and testing stages.
3. Most studies used a combination of datasets from different rocks, including granite, sandstone, limestone, mudstone, marble, and shale, to predict UCS and E. The combination of data from different rocks makes it a challenge for the AI models to identify which specific rock UCS and E have been predicted. Therefore, it is suggested that more work should be based on a particular rock type.

10 Conclusions

This systematic literature review evaluates the position of AI algorithms in enhancing computation and predictions of rock strength and deformation in engineering rock mechanics, emphasising their importance and capabilities in dealing with the complicated behaviour of rock. Additionally, the review gives a thorough assessment of different AI methods utilised in the realms of rock engineering. The main summary emerges regarding the recent status and scope of AI models in rock mechanics:

1. The review demonstrated that AI models can predict UCS and E with high accuracy and minimum errors. Additionally, the study reveals that the ANN and ANN-based algorithms were noticed to be the most widely used in predicting UCS and E. Other AI algorithms, such as RF, SVM, ANFIS, and XGBoost, have likewise proven their potential in forecasting UCS and E of rocks.
2. ANN, ANFIS, and RF-based algorithms per the study collectively constitute substantial proportions of 23.4%, 11.3%, and 8.4%, respectively, for the past decades, demonstrating their usefulness in predicting UCS and E. AI models are used to forecast UCS and E, with a reliability rate of 57.3%, demonstrating robustness in data handling and competence.
3. This systematic literature review catalogues the usage of AI algorithms in predicting UCS strength and E into three different themes: studies that (1) combined physical and mechanical properties, (2) mechanical, and (3) physical properties with AI models. It is observed that a combination of physical and mechanical properties as an input parameter is the most utilised for predicting UCS and E, followed by mechanical and physical properties.
4. High-quality data are essential for accurate AI learning, assessment, and prediction. Prioritising data generation, curation, and management, especially for practical rock engineering tasks, is crucial for the precise application of AI algorithms in predicting UCS and E of rock.
5. In general, AI algorithms show promising potential in estimating the UCS and E of rocks. However, AI models are also associated with some limitations. Therefore, care must be taken when selecting a model from the literature to perform a particular task.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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